# 2

# GOLF

# FUND

# RAIS

# ER

PRO

# Phase 2

# Golf

# er

# Sco
# ring

# App

# +

# QR Tra

# nsfe

# r

# *Expo Mobile App · Offline SQLite ·*

# QR Scorecard · Public Leaderboard

# Golf Scramble Made Easy · Implement

# ation Design Series · v1.0

# 1. Phase 2

# Goal

# &

# Scope

# Phase 2 gives

# golfers their own iOS and Android scoring

# app. Score entry moves from the

organizer's tablet to each team's phone.

# The public leaderboard goes live on the web. QR

# Scorecard Transfer allows zero-inter net score

collection at round end. All Phase 1 admin

capabilities remain and are enhanced.

| | |
|---|---|
| **Timeline** | 8–10 weeks |

| | |
|---|---|
| **New App** | apps/mobile (Expo SDK 55) |

| | |
|---|---|
| **Updated A** | apps/web (Next.js public |

| | |
|---|---|
| **pp s** | leader board) , apps/a pi |

| | |
|---|---|
| | (new endpoints) |
| **Key** | expo-sqlite |

| | |
|---|---|
| **Libraries** | v14, expo-camera, expo-backgro |

| | |
|---|---|
| | und-ta sk, @reac t-navig |

| | |
|---|---|
| | ation/n ative |
| **Pr er eq** | Phase 1 compl |

| | |
|---|---|
| **ui sit e** | ete. All API endpoi nts from |

| | |
|---|---|
| | Phase 1 are used by the |

| | |
|---|---|
| | mobile app. |

# 2. Expo Mobi

# le

# App

# Setup

# 2.1 Project Bootstrap

```
cd
apps/mob
ile
npx
create-e
```

```
xpo-app
--templa
te tabs
# In
app.json
:
```

```
#
"newArch
Enabled"
: true
←
mandator
```

```
y
#
"plugins
":
["expo-s
qlite",
```

```
"expo-ca
mera",
"expo-se
cure-sto
re",#
"expo-ba
```

```
ckground
-fetch",
"expo-no
tificati
ons"]
```

# 2.2 App Screen

# Structure

## app/

```
(auth)/

join.tsx
# Scan
event QR
or enter
```

event code manually

(scoring)/

```
_layout.
tsx
# Tab
navigato
r:
```

# Scorecard | Leaderboard | Team

# scorecard.tsx

# Hole-by-hole score

```
entry

leaderboard.tsx
# Live standing
```

```
s (polls
/leaderb
oard/pub
lic)

team.tsx
```

```
# Team
card:
roster,
tee
time,
hole
```

```
assignment

 (round-end)/
```

sync.tsx

# Aggressive sync screen at round

```
end
qr-trans
fer.tsx
#
Generate
QR for
```

```
admin
scan
```

# 3. Scor

# e Entry UI

# 3.1 Scorecard Screen

- Hole navigator: swipe left/right or tap

hole number. Current hole

large and centered.

- Gross score entry:

# two large ± buttons (minimum 56pt touch

targets — golf glove requirement).

# Score displayed as large centered number.

- To-par running total: displayed at top at

all times. E.g. '-3 thru 9'. Updates instantly

on each score change.

- Player shot

tracking (optional): below the gross score, a

row of player avatar buttons. Scorekee

per taps which player's drive was used

each hole. Stored in scores.player_sho

ts

JSONB.

- Additiona

l fields

(collapsib

le): putts count. Shown/hidden per

event config.

- Previous hole edit: swipe

right from hole 1 brings back the last entry.

# Tap any hole number in the

hole bar to jump. Hole sponsor: if the

hole has a sponsor, show sponsor

# logo at top of hole screen with

name. Guaranteed impression.

# 3.2 Score Submi

# ssion Flow

1. Scorekeeper

changes gross score for hole N.

2. Score immediately written to SQLite

# pending_scores table (< 1ms,

3. never fails). Background sync task

picks it up and POSTs to PATCH /api/v1/sc

ores/{teamId}/{holeNumber}.

4. On success: record marked synced_a

t in SQLite. On failure: retry with

5. exponential backoff. API recalculat

es
leaderbo
ard
standings
. Public

# leaderboard reflects change on next

poll (Phase 2) or immediately via

# SignalR push (Phase 3).

# 4. Offline-Fi

# rst

# Archi

# tectu

# re

# 4.1 SQLite Schema

# (expo-sqlite v14,

# WAL mode)

```
-- Event cache
```

(downloaded at join, never changes during

```
round)
CREATE
TABLE
event_ca
che (key
TEXT
```

```
PRIMARY
KEY,
value
TEXT);
--
Stores:
```

event
config,
course
pars,
hole
handicap

```
_index,
team
roster,
cup_loca
tion
(Phase
```

6)

--

Pending scores (written

immediately, synced when connected)

```
CREATE
TABLE
pending_
scores (
    id
TEXT
```

```
PRIMARY
KEY,

team_id
TEXT NOT
NULL,
```

```
hole_number INTEGER NOT NULL,
```

```
gross_sc
ore
INTEGER
NOT
NULL,
```

```
putts
INTEGER,

player_s
hots
TEXT,
```

```
-- JSON

created_
at TEXT
NOT
NULL,
```

```
synced_at TEXT,
-- NULL = not yet
```

synced

sync_attempts INTEGER DEFAULT

```
0
);

--
Leaderbo
ard
```

```
cache
(refresh
ed on
successf
ul
sync)CRE
```

```
ATE
TABLE
leaderbo
ard_cach
e (key
TEXT
```

```
PRIMARY
KEY,
value
TEXT,
cached_a
t TEXT);
```

# 4.2 Connectivity Tier

# Detection

- Full (LTE/5G

WiFi):
sync
immediat
ely on
score

# write. Real-time leaderboard

polling active.

- Degraded (3G): sync on

write, but accept higher latency. Leaderbo

ard polling interval extended to 60s.

- Offline: scores queue in SQLite. Leaderbo

ard shows cached data with 'Last

# updated X min ago' banner.

# Intermittent: aggressive retry — backgrou

nd task runs every 30s. On reconnec

t, drains the full pending_scores queue.

# 4·3 Pre-Flight

# Checklist

## Before the round

# starts, the app displays a pre-flight checklist

that must be completed. The scorecard

# is locked until all items are green.

- ✓ Event joined and event_cache

populate
d

- ✓ Team
roster
confirme

d (all players visible on team screen)

- ✓ Tee time / starting hole

confirme

d

- ✓

Network

connectiv

ity check passed OR 'Start in Offline Mode'

explicitly confirmed

- ✓ Battery

level > 30% (warning only, not blocking)

# 5. QR Scor ecar

# d Tran sfer

# At round end, if internet sync is incomplet

e, the admin opens a QR collection

screen and each team's scorekeeper

generates a QR code from their phone.

# Admin scans each team in sequence.

# No internet required for either device

# during the transfer.

# 5.1 Scorek

# eeper Side

Scorekee per taps

'Round Complete' → 'Generate

# Scorecard QR' App compresses

# 18-hole scorecard to JSON, signs

with HMAC-SHA256 (key = event_co

de + team_id), Base64-encodes, renders

# as QR code Standard 18-hole payload

# ~900 bytes — fits single QR at error

correction level M (≤ 1,273 bytes)

- If payload > 1,273 bytes: app

auto-splits into Part 1 of 2 and Part 2 of

# 2 QR codes

- Screen brightness

maximized automatically. QR stays on

screen until admin confirms receipt.

# 5.2 QR Payload

# Structure

```
{
    "v":
```

```
1,
//
schema
version
  "ec":
"GLFGIVE
```

```
4",
//
event_co
de
  "tid":
"team-uu
```

```
id",
//
team_id
  "tn":
"Team
Birdie",
```

```
//
team_name
  "did":
"device-
uuid",
```

```
//
scorekee
per
device
ID
  "ts":
```

```
1710000000,
// Unix
timestamp of
generati
```

```
on
    "sig":
"abc123.
..",
//
HMAC-SHA
```

```
256
signatur
e

"scores"
: [
```

```
// 18 elements, null if hole not played
```

```
      {
"h": 1,
"g": 5,
"p": 2
},
      {
```

```
"h": 2,
"g": 4,
"p": 1
},

...

]}
```

# 5·3 Admin Collect

# ion Side

- Admin opens

# 'Collect QR Scores' on their device

(tablet/phone). Camera activates.

- Camera stays open continuously —

chime on each successful scan. Running

counter: 'Collected: 18 of 25 teams'.

- POST /api/v1/events/{id}/scores/qr-collect

—

validates HMAC signature, imports

scores, marks source = qr_transfer.

# Tampered payload (invalid signature

# ):

# rejected with error toast.

# Admin

# manually enters scores for that team.

- Admin dashboard shows 'Pending Sync'

counter.

Blocks final leaderboard

publish until all teams confirmed or

explicitly waived.

# 6. Public

# Leaderboard

# (Next.js)

- Route: /e/{orgSl

ug}/{eventCode}/scores — public, no login

required. Phase 1 landing page updated

with 'Watch Live Scores' button

# linking here. SSR initial load:

# page renders with current standings

# server-side for fast first paint and search

engine indexing. Client-side polling: after

# hydration, polls GET /api/v1/pub/event

s/{eventCode}/leaderboard every 15 seconds.

# Updates standings in place without

full reload. Upgrade in Phase 3: polling

# replaced by SignalR push

subscripti
on.
- TV
mode:
/e/{orgSl

ug}/{eventCode}/scores?tv=1 — large

font, auto-scrolling, reduced UI

chrome.

Designed for 55"+ clubhous

e displays.

- Hole challenge tab:

shows current leaders per challenge

# (closest to pin distance, longest drive

distance)

. Updates on each poll.

- Donation thermometer: total raised shown in

sidebar. Updates on poll. Sponsor banner:

# Title sponsor in header. Gold

sponsors in rotating banner. Hole

sponsor shown when viewing per-hole

# breakdown.

# 7. Phase 2

# New API

# Endpoints

# IEndpoint Description

| | | |
|---|---|---|
| C<br>C | | |
| F<br>C | /ev<br>ent<br>s/{i | Golfe<br>r<br>joins |

| | | |
|---|---|---|
| s | d}/j<br>oin | event<br>via<br>QR<br>token<br>or |

| | | |
|---|---|---|
| | | event_code. Returns |

| | | |
|---|---|---|
| | | team card + full event_cac |

| | | |
|---|---|---|
| | | he paylo ad for SQLit e |

| | | |
|---|---|---|
| | | down load. |
| F C | /sy nc/ sc | Batch offlin e |

| | | |
|---|---|---|
| s | ore | score |
| l | s | sync. |
| | | Body: |
| | | array |
| | | of |

| | | |
|---|---|---|
| | | pending_scores.Retur |

| | | |
|---|---|---|
| | | ns confli ct list. |
| F C | /ev ent | Admi n |

| | | |
|---|---|---|
| s | s/{id}/scores/q | submits QR-scanned |

| | | |
|---|---|---|
| | r-c<br>oll<br>ect | score<br>card<br>paylo<br>ad.<br>Valid |

| | | |
|---|---|---|
| | | ates<br>HMA<br>C.<br>Impor<br>ts |

| | | |
|---|---|---|
| | | scores. |
| C<br>E<br>T | /pub/ev | Public leade |

| | | |
|---|---|---|
| | nts /{code}/leaderboard/{code}/standings, thru, | |

| | | |
|---|---|---|
| | der bo ard | to-pa r, spon sor data. |

| | | |
|---|---|---|
| | | No auth. |
| C<br>E<br>T | /pu<br>b/e<br>ve | Publi<br>c<br>chall |

| | | |
|---|---|---|
| | nts /{code}/ch | enge leaderboard. No auth. |

| | | |
|---|---|---|
| | all<br>en<br>ge<br>s | |

| | | |
|---|---|---|
| C | /pu | Publi |
| E | b/e | c |
| T | ve | fundr |
| | nts | aisin |
| | /{c | g |

| | | |
|---|---|---|
| | od<br>e}/f<br>un<br>dra | total.<br>No<br>auth. |

| | | |
|---|---|---|
| | isi<br>ng | |

# 8. Phase 2

# End-to-End

# Flow Summary

6. Organizer creates event (Phase 1).

# Golfers register online (Phase 1).

7. Morning of event: organizer opens admin

dashboar
d.
Checks
pre-flight.
Generate

s QR print kit and paper backups.

8. Golfers arrive. Admin scans QR

9. codes for check-in. Shotgun start. Each

team's scorekee per opens mobile

# app, scans event join QR.

# 10. Pre-flight checklist runs. Scorecard

11. unlocked when all green. Golfers play.

# Scorekeeper enters hole-by-hole

scores.

Scores sync to server when

12. connected. Public leaderboard polls

for updates every 15 seconds. Friends,

family, donors watch standings from

13. anywhere. Round ends. Admin

# opens 'Collect QR Scores'. Walks

# from team to team at 18th green.

# Each team generates scorecar

d QR. Admin scans. 25 teams collected

14. in ~10 minutes. Admin reviews any

conflicts. Confirms final standings. Marks

event complete

.

Thank-you emails

fire automatically.